

# Private Service Access and Traffic Inspection

Endpoints, PrivateLink, and putting a firewall in the middle of everything.

A gateway endpoint is a **route**. An interface endpoint is an **address**. Almost every difference follows from that.

# Gateway or interface endpoint?

## GATEWAY ENDPOINT

### A route table entry

- S3 and DynamoDB only
- No hourly or per-GB charge
- Only works inside the VPC
- Not reachable from on-premises

## INTERFACE ENDPOINT

### An ENI with a private IP

- Almost every AWS service
- Per AZ per hour, plus per GB
- Reachable over peering and TGW
- Works from on-premises

## THE TIEBREAKER

On-premises, a peered VPC or a Transit Gateway in the question eliminates the gateway endpoint immediately.

# Gateway endpoint details

- ◆ **It adds a prefix list to a route table.** Choose which route tables, and the subnets using them are covered.
- ◆ **Free.** It also removes NAT gateway processing charges for that traffic, which is often the real saving.
- ◆ Not reachable from on-premises, a peered VPC, or over a Transit Gateway. It is a local route.
- ◆ An **endpoint policy** limits what can be done through it, independently of IAM.

## ON THE EXAM

The saving is rarely the endpoint fee, which is zero anyway. It is the NAT gateway processing charge you stop paying.

# Interface endpoint details

- ◆ **An ENI per subnet you select**, each with a private IP. You pay per ENI per hour and per GB.
- ◆ **Private DNS** makes the normal service hostname resolve to the endpoint, so applications need no change.
- ◆ It has a **security group**, which a gateway endpoint does not.
- ◆ **Cross-Region PrivateLink** now reaches AWS services in other Regions, which was impossible until late 2025.

## ON THE EXAM

Private DNS is why applications never notice. Turn it off and clients keep resolving the public endpoint instead.

# Locking a service to your network

- ◆ An **endpoint policy** controls what can pass through the endpoint. It grants nothing on its own.
- ◆ **aws:SourceVpce** in a bucket policy means: only requests arriving through this endpoint.
- ◆ **aws:SourceVpc** is the broader form, restricting to a whole VPC rather than one endpoint.
- ◆ An endpoint alone does not stop anyone reaching the bucket publicly. The resource policy does that.

## ON THE EXAM

Creating an endpoint changes the path traffic takes. Only a resource policy condition makes it the only permitted path.

# Publishing your own service with PrivateLink

- ◆ **You create an endpoint service** in front of a Network Load Balancer or Gateway Load Balancer.
- ◆ Consumers create an interface endpoint. **You accept or reject each connection**, and name permitted principals.
- ◆ **Address space never has to agree**, which is why it survives overlapping CIDRs.
- ◆ It is unidirectional. The consumer reaches you; you do not reach the consumer.

## ON THE EXAM

One service, one direction, no shared routing. That combination is what makes PrivateLink the answer for SaaS and acquisitions.

# Two newer endpoint types, and an honest scope note

- ◆ **Resource endpoints** reach an RDS instance, an IP, or a domain in another VPC or on-premises, with **no load balancer**.
- ◆ **Service network endpoints** reach many shared services and resources through one endpoint.
- ◆ Both are **Amazon VPC Lattice** features, and **VPC Lattice is not on the exam's in-scope list**.
- ◆ Know they exist. Answer with an NLB-fronted endpoint service, which is what the exam expects.

## ON THE EXAM

This is a real gap between current AWS and the exam. In production these often beat an endpoint service; on the exam they are not the answer.

# Gateway Load Balancer

- ◆ **For third-party appliances:** firewalls, intrusion prevention, deep packet inspection.
- ◆ **Encapsulates traffic in GENEVE on port 6081**, so the appliance sees the original packet unaltered.
- ◆ **Transparent.** Traffic is routed to a GWLB endpoint, inspected, and continues. No address changes.
- ◆ Scales the appliance fleet and health checks it, so one dead firewall does not black-hole traffic.

## ON THE EXAM

GWLB is the answer whenever a scenario names a commercial appliance. Network Firewall is the answer when it does not.

# Centralised inspection and egress

01

## Spoke VPC

Default route points at the Transit Gateway.

02

## Transit Gateway

A spoke route table sends everything to the inspection VPC.

03

## Inspect

GWLB endpoint or Network Firewall examines the traffic.

04

## Egress

One NAT gateway set serves every VPC.

### ON THE EXAM

Enable appliance mode on the inspection VPC attachment, or asymmetric routing breaks stateful inspection. Module six.

# Network Firewall or Gateway Load Balancer?

## AWS NETWORK FIREWALL

### Managed, AWS-operated

- No instances to run or patch
- Suricata-compatible rules
- Domain filtering and IPS
- Scales without your involvement

## GATEWAY LOAD BALANCER

### Your chosen appliance

- Run a specific vendor's product
- Reuse existing rules and expertise
- You own patching and capacity
- Needed when the vendor is named

## THE TIEBREAKER

A named vendor, or existing rules that must be reused, means GWLB. Anything else favours the managed service.

# Reading the scenario

## THE SCENARIO SAYS

Large NAT charges for S3 or DynamoDB traffic

On-premises needs private access to an AWS service

The bucket must only be reachable from our VPC

Expose one service to customers, overlapping CIDRs

Inspect with a named vendor appliance

Endpoint costs repeated in dozens of VPCs

## IT IS POINTING AT

Gateway endpoint

Interface endpoint

aws:SourceVpce in the bucket policy

Endpoint service behind an NLB

Gateway Load Balancer

Centralise, plus a private hosted zone

## ON THE EXAM

Ask where the caller sits. Inside the VPC allows a gateway endpoint; anywhere else forces an interface endpoint.

Meridian Freight runs a nightly analytics job on EC2 in private subnets that reads about 40 TB per month from Amazon S3 in the same Region. Traffic reaches S3 through a NAT gateway.

Finance has queried the NAT gateway line, which is dominated by data processing charges rather than hours.

### What should the solutions architect recommend?

- ☐ A Create a gateway endpoint for S3 and add it to the private subnets' route tables
- ☐ B Create an interface endpoint for S3 in each Availability Zone
- ☐ C Move the instances to public subnets with Elastic IPs
- ☐ D Replace the NAT gateway with NAT instances

### ◆ PAUSE AND DECIDE

## What should the solutions architect recommend?

- A** Gateway endpoint  
*Free, and it takes 40 TB a month off the NAT gateway entirely. Same Region, same VPC, so every condition is met.*
- B** Interface endpoint  
*Works, and charges per ENI per hour plus per GB. You would pay for 40 TB when a gateway endpoint carries it for nothing.*
- C** Public subnets  
*Removes the NAT charge by putting analytics instances on the internet. A security regression to solve a billing problem.*
- D** NAT instances  
*Still processes the same 40 TB, now on instances you patch and scale. Module six's false economy, repeated.*

### THE SIGNAL

Same Region, same VPC, S3. All three conditions for the free option hold, so paying per GB would be a choice.

The same company now needs its on-premises backup software to write directly to Amazon S3 across the existing Direct Connect, without traversing the internet.

An engineer suggests reusing the gateway endpoint created for the analytics workload.

### What should the solutions architect advise?

- ☐ A Reuse the gateway endpoint and add a route on-premises
- ☐ B Create an interface endpoint for S3; gateway endpoints cannot be reached from on-premises
- ☐ C Use a public VIF and rely on the bucket policy for security
- ☐ D Deploy a proxy fleet in the VPC that forwards requests to the gateway endpoint

### ◆ PAUSE AND DECIDE

## What should the solutions architect advise?

**A** Reuse the gateway endpoint

*A gateway endpoint is a route inside one VPC. On-premises routers have no way to use somebody else's route table.*

**C** Public VIF

*Genuinely reaches S3 over the circuit rather than the internet, and it uses public addressing, which the requirement rules out.*

**B** Interface endpoint

*It has a private IP that on-premises can reach over Direct Connect, which is exactly what a gateway endpoint lacks.*

**D** Proxy fleet

*Builds and operates infrastructure to emulate a service AWS already sells, and inserts a bottleneck into a backup path.*

### THE SIGNAL

One constraint changed from scenario one: the caller moved outside the VPC. That alone flips a free answer to a paid one.

An auditor requires that a bucket holding customer records can be reached only from the company's own VPCs, and never from the internet, even with valid credentials.

The team has created interface endpoints in every VPC and believes the requirement is met.

### What should the solutions architect tell them?

- ☐ A The requirement is met; endpoint traffic does not traverse the internet
- ☐ B Add a bucket policy denying requests unless aws:SourceVpce matches the approved endpoints
- ☐ C Enable S3 Block Public Access on the bucket
- ☐ D Add an endpoint policy on each interface endpoint allowing only that bucket

### ◆ PAUSE AND DECIDE

## What should the solutions architect tell them?

- A

**Already met**  
*Endpoints change the path their traffic takes. The bucket's public endpoint is untouched and still answers anyone with credentials.*
- B

**Deny unless aws:SourceVpce**  
*Makes the endpoint the only permitted path. Requests arriving any other way are denied regardless of credentials.*
- C

**Block Public Access**  
*Blocks anonymous and public access. A stolen credential is neither, and the scenario says even with valid credentials.*
- D

**Endpoint policies**  
*Limits what may pass through the endpoint. It says nothing about requests that never use the endpoint.*

### THE SIGNAL

Creating a path is not restricting a path. Only a condition on the resource policy closes the others.

A software company sells a fraud-scoring API. Customers run it from their own AWS accounts, many of which use 10.0.0.0/16, as does the provider. Customers must not be able to reach anything else in the provider's VPC.

The current design asks each customer to accept a VPC peering connection.

### What should the solutions architect recommend?

- ☐ A Continue with peering and use security groups to restrict access
- ☐ B Create a PrivateLink endpoint service behind a Network Load Balancer
- ☐ C Publish the API on the internet behind an Application Load Balancer with IP allow-lists
- ☐ D Attach every customer to a shared Transit Gateway

### ◆ PAUSE AND DECIDE

## What should the solutions architect recommend?

- A

**Peering plus security groups**  
*Peering refuses overlapping CIDRs outright, so most customers cannot connect at all. Security groups never get a chance to help.*
- B

**PrivateLink endpoint service**  
*Exposes one service, tolerates identical address space, and gives the provider no route into customer networks.*
- C

**Public ALB with allow-lists**  
*Puts a fraud API on the internet and asks customers to maintain address lists that change whenever they scale.*
- D

**Shared Transit Gateway**  
*Routing again, so overlapping CIDRs still fail, and it would join customers' networks to each other.*

### THE SIGNAL

Overlapping CIDRs plus one service plus mutual distrust is the PrivateLink signature. All three point the same way.

A bank must inspect all traffic between its VPCs using the same commercial firewall product it runs on-premises, because its rule base and its regulators' approvals are tied to that vendor.

The proposed design routes inter-VPC traffic through an inspection VPC with the appliances spread across three Availability Zones.

### What should the design use, and what must be enabled?

- ☐ A AWS Network Firewall in the inspection VPC, with appliance mode on its attachment
- ☐ B Gateway Load Balancer fronting the appliances, with appliance mode on the inspection VPC attachment
- ☐ C A Network Load Balancer fronting the appliances, with cross-zone load balancing enabled
- ☐ D An Application Load Balancer fronting the appliances, with sticky sessions

#### ◆ PAUSE AND DECIDE

## What should the design use, and what must be enabled?

**A** Network Firewall

*Excellent and managed, and it is not the named vendor. The regulatory approval and the rule base both attach to a specific product.*

**C** NLB, cross-zone

*An NLB is a load balancer, not a transparent bump in the wire. Traffic would need to be addressed to it rather than routed through.*

**B** GWLB plus appliance mode

*GWLB runs the chosen vendor transparently over GENEVE, and appliance mode keeps both directions of a flow in one AZ.*

**D** ALB, sticky sessions

*Layer 7 only, so it cannot inspect arbitrary protocols, and it terminates connections rather than passing them through.*

### THE SIGNAL

A named vendor forces GWLB. Stateful appliances across AZs forces appliance mode. The question tests both at once.

A company has 55 VPCs, each with interface endpoints for Secrets Manager, Systems Manager, ECR and CloudWatch Logs, in three Availability Zones. The endpoint charges have become a noticeable monthly line.

All VPCs already attach to a Transit Gateway.

### What should the solutions architect recommend?

- ☐ A Reduce each VPC's endpoints to a single Availability Zone
- ☐ B Centralise the endpoints in a shared services VPC, and use a private hosted zone shared across accounts
- ☐ C Remove the endpoints and route the traffic through NAT gateways instead
- ☐ D Replace the interface endpoints with gateway endpoints

### ◆ PAUSE AND DECIDE

## What should the solutions architect recommend?

- A** One AZ per VPC  
*Cuts the bill by two thirds and creates a single point of failure in every VPC for secrets and logging.*
- B** Centralise plus private hosted zone  
*660 endpoints become 12. The TGW already provides the path, and a shared private hosted zone points clients at the central ENIs.*
- C** Back to NAT gateways  
*Trades endpoint charges for NAT processing charges and puts service traffic across the internet, losing the privacy the endpoints bought.*
- D** Gateway endpoints  
*Only S3 and DynamoDB have them. None of these four services does, so the option is not available.*

### THE SIGNAL

Interface endpoints are reachable over a Transit Gateway, which is what makes centralisation possible. Gateway endpoints never could be.

# What to carry out of this module

- ◆ **Gateway endpoint is a route; interface endpoint is an address.** Every difference follows.
- ◆ **On-premises, peering or a TGW rules out gateway endpoints.** They cover only S3 and DynamoDB.
- ◆ **Creating an endpoint is not restricting access.** Only `aws:SourceVpce` closes the other paths.
- ◆ A named vendor appliance means **GWLB**. Stateful inspection across AZs means **appliance mode**.

## ON THE EXAM

Ask where the caller sits, then whether the requirement is a new path or a closed one. Those two questions cover most of this area.